Exercise 4

Joins, UNION, Filtering and aggregates

1) SQL Joins

a) SELECT Employee ID,
Firstname,
Lastname,
Department,
Salary,
ProjectID,
Projectname,
Budget,
Status

from employees As A
Inner JOIN projects As B
ON A.employeeID = B.EmployeeID;

EmployeeID	Firstname	Last names	Departments	Salary	ProjectID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Complete
2	Alice	Smith	HR	60 000	102	Employee training	50 000	Ongoing
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	105	Market expansion	65 000	Complete
6	Michael	Clark	Finance	80 000	106	Risk management	80 000	Pending

3) SELECT employee ID,
First name,
Lastname,
Department,
Salary,
Project ID,
Projectname,
Budget,
Status

FROM employees As A
LEFT Join projects A B
ON A. employee ID = B. employee ID;

employeeID	Firstname	Lastname	Department	Salary	ProjectID	Projectname	Budget	Status
1	John	Doe	IT	70 000	103	Cybersecu	75000	Pending
2	Alice	Smith	HR	60 000	102	Employee Tr	50 000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Ana	90000	Ongoing
4	David	Brown	IT	72 000	null	Null	null	null
5	Emma	Wilson	Sales	65000	105	Market Eve	65000	Complete
6	Michael	Clarke	Finance	80 000	106	Risk Mana	80 000	Pending
1	John	Doe	IT	70 000	101	AI Develo	100 000	Compl

c) SELEC Project ID,
Projectname,
Budject,
Status,
Employee ID,
Firstname,
Lastname,
Department,
Salary
FROM project AS B
Right Join employees AS A
ON A.employeesID = B.employees ID;

Project ID	Project Name	Budget	Status	Employee ID	Firstname	Lastname	Department	Salary
103	CyberSecurity Audit	25000	Pending	1	John	Doe	IT	70 000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60 000
104	Financial Analyst	90000	Ongoing	3	Bob	Johnson	Finance	75000
NULL	NULL	NULL	NULL	4	Daniel	Brown	IT	72000
105	Market expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk management	80 000	Pending	6	Michael	Clark	Finance	80 000
101	AI Development	100 000	Completed	1	John	Doe	IT	70 000

d) SELECT employee ID,
Firstname,
Lastname,
Department,
Salary;
Project ID,
Projetname,
Budget,
Status

FROM employees AS A
Full OUTER JOIN projets AS B
ON A. employeeid = B. employee ID;

Employee ID	Firstname	Lastname	Department	Salary	Project ID	Projectname	Budget	Status
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90 000	On going
4	David	Brown	IT	72000	NULL	Null	null	NULL
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50 000	On going
5	Emma	Wilson	Sales	65000	105	Market expansion	65 000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
1	John	Doe	IT	70000	101	HR Development	100 000	Completed

3 a) SELECT employee ID,
Firstname,
Lastname,
Department,
Salary
FROM employees
WHERE salary > 70 000;

EmployeeID	Firstname	Lastname	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance.	80 000

b) SELECT employee ID,
Firstname,
Lastname,
Department,
Salary
FROM employees
WHERE Department = IT or Department = Finance;

EmployeeID	Firstname	Lastname	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

c) SELECT projectID,
Projectname,
Budget,
Status

FROM projects
WHERE status != completed;

Project ID	Projectname	Budget	Status
102	Employee training	50000	Ongoing
103	CyberSecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80000	Pending

D) SELECT projectID,
projectname,
Budget,
Status

FROM employees
WHERE Budget > 70 000
AND status != completed;

Project ID	Project name	Budget	Status
103	CyberSecurity Au	75 000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	8 0000	Pending

E) SELECT Employee ID,
Firstname,
Lastname,
Department,
Salary,
City
FROM Employees
WHERE city = "New York" or City = "Toronto"
ORDER BY salary DESC;

Employee ID	Firstname	Lastname	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75 000	Toronto
1	John	Doe	IT	70 000	New York

F) SELECT EmployeeID,
Firstname,
Lastname,
Department,
Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

Employee ID	Firstname	Lastname	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

4) A)
```
SELECT Department,
       Sum(salary) As Total Salary
FROM Employees
GROUP BY Department,
ORDER BY total Salary DESC;
```

Department	Total Salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60000

B)
```
SELECT city,
       AVG (Salary) As Average Salary
FROM Employees
GROUP BY city,
HAVING average salary > 65000;
```

City	Average Salary
New York	75 000
Toronto	75000
London	72 000

C)
```
SELECT Department,
       COUNT (EmployeeID) As EmployeeCount
FROM Employees
HAVING employeeCount > 1
```

Department	EmployeeCount
IT	2
Finance	2

d) SELECT status
COUNT (projectID) AS ProjectCount
FROM projects
GROUP BY status,
HAVING projectCount > 1;

Status	Project Count
Completed	2
Ongoing	2
Pending	2

E) SELECT EmployeeID
First name,
Lastname
SUM (budget) AS TOTAL projectbudget
FROM employee AS A
JOIN projects AS B
ON A. employee ID. = B. employeeID
GROUP BY A. employee, A firstname, A lastname;
HAVING total projectbudget > 150 000;

Employee ID	First Name	Lastname	Total project Budget
1	John	Doe	175 000